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RESEARCH ARTICLE

Raw Camera Data Object Detectors: An Optimisation for Automotive Video Processing and Transmission

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ABSTRACT Whilst Deep Neural Networks (DNNs) have been developing swiftly, most of the research has been focused on videos based on RGB frames. RGB data has been traditionally optimised for human vision and is a highly re-elaborated and interpolated version of the collected raw data. In fact, the sensor collects the light intensity value per pixel, but an RGB frame contains 3 values, for red, green, and blue colour channels. This conversion to RGB requires computational resource, time, power, and increases by a factor of three the amount of output data. This work investigates DNN based detection using (for training and evaluation) Bayer frames, generated from a benchmarking automotive dataset (*i.e.* KITTI dataset). A Deep Neural Network (DNN) is deployed in an unmodified form, and also modified to accept only single channel frames, such as Bayer frames. Eleven different re-trained versions of the DNN are produced, and cross-evaluated across different data formats. The results demonstrate that the selected DNN has the same accuracy when evaluating RGB or Bayer data, without significant degradation in the perception (the variation of the Average Precision is $<1\%$). Moreover, the colour filter array position and the colour correction matrix do not seem to contribute significantly to the DNN performance. This work demonstrates that Bayer data can be used for object detection in automotive without significant perception performance loss, allowing for more efficient sensing-perception systems.

INDEX TERMS Bayer data, object detection, perception sensors, assisted and automated driving, intelligent vehicles.

I. INTRODUCTION

With the advancement of computer hardware technology, deep learning-based artificial intelligence technologies are in rapid development, and are used in a wide range of applications, including assisted and automated driving (AAD) functions [1]. The Society of Automotive Engineers (SAE) J3016 standard defines six levels of driving automation (L0-L5) [2]. As functions on vehicles reach higher levels of automation (L3-L5), the ability to sense and make decisions based on the external environment becomes an increasingly essential capability. As a foundation for path planning, behavioural decisions, and motion control, environmental perception is a key research topic in academia and industry [3]. The detection of traffic actors such as vehicles and pedestrians, and the implementation of real-time vehicle perception of the road conditions are important for the prevention of common types of traffic accident [4], [5].

Deep neural network (DNN) methods are well established techniques for detecting and classifying objects, and there is a

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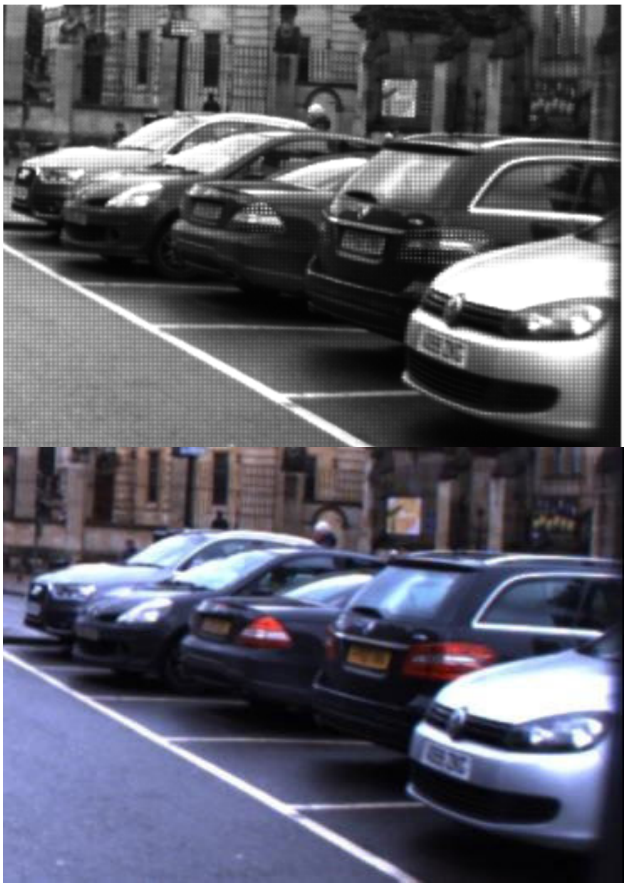


FIGURE 1. A frame from the Oxford Robocar dataset [17]. Top is the raw frame from the dataset, bottom is the same frame after ISP processing and conversion in 3 colour channels.

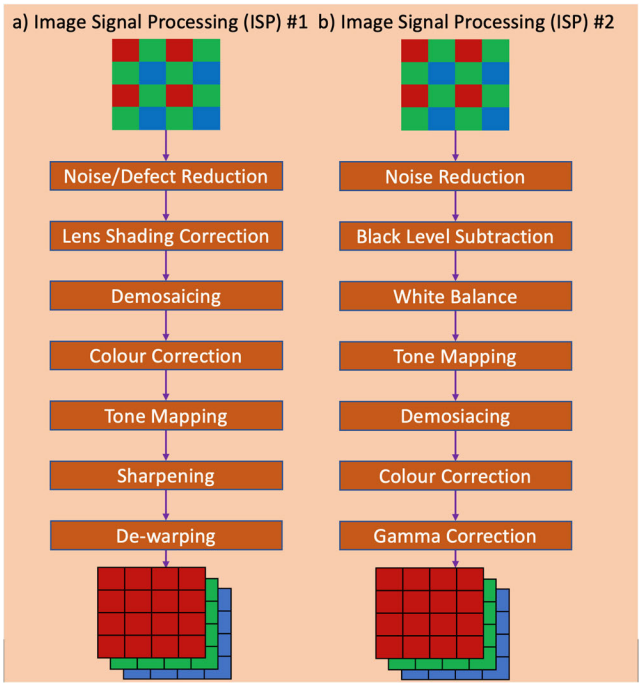


FIGURE 2. a) Some fundamental processing steps in a generic colour pipeline in an imaging sensor, and b) the ISP blocks considered in [37]. In both cases, the output is a 3 colour channel frame with the same resolution of the input Bayer data.

rapidly growing body of work related to their use for detection of road stakeholders [6]. The R-CNN and YOLO series are the most commonly used DNNs for object detection tasks, but there is a trade-off between detection accuracy and detection speed [7]. Until recently, most of the DNNs have been based (*i.e.* trained and tested) on frames with three colour channels, RGB (red, green, blue). In automotive, the RGB inputs to DNNs are the frames produced by HDR video cameras, and in turn they are a processed version of the captured *raw sensor data*. The term ''raw'' is often misused in literature to define data that are captured and post-processed by the sensor, *i.e.* unrectified images with RGB colours per pixel [8], [9]. In reality, the raw data corresponds to a value of light intensity collected per pixel through the sensor's colour filter array (CFA), as shown in Fig. 1. Traditionally the CFA has been in the format of a R-G-G-B 2×2 repeated pixel matrix and optimised for human vision [10]. The conversion into RGB colour channels, through the colour pipeline and ISP (image signal processing) in the sensor, has been historically created to produce frames looking pleasant and realistic to human viewers. However, this processing and manipulation might be not needed for machine learning and DNN-based perception, and this paper aims to explore if *raw* or *Bayer* data (*i.e.* one intensity value per pixel or one intensity value and the specific colour of the filter on the pixel) can be used for perception without degrading DNN performance. Moreover, the use of raw data will reduce, roughly by a factor of three, the size of the data to be transmitted into the processing algorithms (only one value per pixel instead of three), and will decrease the processing on chip [11]. Recent work in different fields has been focusing on raw image camera consumption, *e.g.* [12], [13], [14], [15], and [16], and it is further discussed in Sec.II.

The ISP pipeline includes different manipulations of the raw data, including noise reduction, black level andwhite balance correction, colour balancing, gamma correction, dead pixels concealment, *etc*. Two general and similar diagrams for ISP pipeline are shown in Fig. 2. There are several different ways of implementing an ISP and they are outside the scope of this paper. This paper will focus on the contribution of two key steps in the colour pipeline, the *demosaicing* process and the *colour correction matrix*. The raw values captured by the camera are measured based on the light incoming from the real world at the specific position of the sensor pixel matrix (one intensity value per pixel). The demosaicing process then consumes these raw Bayer values to interpolate colour values for the three channels, each channel retaining the same resolution as the sensor pixel matrix. This process is implemented by interpolating the values of neighbouring pixels in the raw matrix, and different algorithms can be applied. The colour correction matrix is used to balance the gains per colour channel to ensure the colour rendering of the frames is realistic (for the human viewer).

The importance of the demosaicing process is further discussed by Chang et al. [18], where the authors discuss the importance to use raw data when considering downstream tasks, namely super-resolution.

A. CONTRIBUTIONS

This paper builds on the considerable and recent work on raw camera data proposing the following innovations.

- Given the scarcity of raw datasets [14], [16], this work proposes different ways of 'inverting' RGB datasets into *'Bayer'* datasets. This step can be key for generating big Bayer datasets from existing datasets for the re-training of state of the art (SOTA) DNN models.
- Considering the combination of raw data with downstream tasks [15], [16], [18], for the first time, the re-created *Bayer* datasets are used to re-train and test a commonly used DNN based object detector.
- On the contrary of most of the existing work, hereby it is proposed an elegant a fair comparison between DNN performance when the DNN is using *Bayer* data versus 'traditional' RGB/grayscale frames.

Overall in alignment with recent work [12], [16], [18], the results show that it is possible to use *single colour channel 'Bayer' data* and to achieve detection performance comparable to the performance with traditional R data. This demonstration enables the reduction of needed processing on the sensor chip, and to transmit less camera data (minimally conditioned) to the vehicle processing unit(s), helping to address the previously presented data conundrum [11], [19].

II. RELATED WORK

This paper builds on the recent work on raw camera data (Sec.II-A) and on unfolding the relationship between DNN performance and ISP (Sec.II-B). The proposed experiments aim at understanding if commonly used DNN architectures (Sec.II-C) can be easily re-used (*e.g.* just by transfer learning) with Bayer frames, or a paradigm shift is needed in Bayer-based DNNs.

A. USE OF RAW DATA IN MACHINE LEARNING

An increasing amount of research has been conducted in the use of raw data, especially fields involving machine learning. Dong et al. have included the use of raw data for pretraining of video to text tasks [12]. Pan et al. presented a denoising network which predicts the noise in a real-world image at the raw level [13]. Song et al. have employed raw image to identify reflections in images using a DNN (CR3Net), allowing for the removal of reflections in images [14].

Specifically to automotive, overexposure of images is a concern and can create hazards. Fu et al. have designed a method using a Channel-Guidance Network (CGNet) to correct over exposure in raw images [15]. This work demonstrated that applying correction to raw data is beneficial to perception performance. In addition, the developed CGNet for the raw image performed better than existing methods when inferring on raw images, showing the importance of developing specific networks for raw [15]. Xu et al. have presented a novel HDR dataset in raw [16]. From this dataset, they have applied different ISP processes and tested the object detection accuracy using YOLOX, demonstrating that ISP impacts the accuracy of detection.

B. OBJECT DETECTION METHODS

Some of the most important targets in traffic scenes are vehicles, pedestrians, and cyclists [20]. There are many studies regarding detection of these objects, and they can be broadly divided into three categories, discussed below: traditional detection methods, traditional machine learning methods, and deep neural network methods. These studies have used RGB frames, but recent work has also started to look into Bayer frames, as reviewed in sec. II-C.

1) TRADITIONAL DETECTION METHODS

These handcrafted object recognition methods, often based on regression, are difficult to apply to a wide range of real-world situations. Accurate results are difficult to obtain when weather conditions change, objects are obstructed or too dark, etc. Furthermore, different targets require different classifiers to be developed and real-time detection is impossible [21], [22], [23], [24], [25].

2) TRADITIONAL MACHINE LEARNING DETECTION METHODS

Traditional machine learning detection algorithms for vehicles generally propose new vehicle-specific features or use other environmental information as an auxiliary detection method. Laopracha et al. employed V-HOG features in combination with SVM kernel functions to detect vehicles, ensuring both accuracy and speed of the overall algorithm [26]. Based on histograms of oriented gradients, Cao et al. developed a vehicle detection system based on the AdaBoost classifier, which can basically meet the requirements of real-time vehicle detection [27]. Similarly, pedestrian detection algorithms have been dominated by the introduction of new features, or multi-feature fusion methods. Bastian et al. presented the second-order aggregate channel features (SOACF) for pedestrian detection [28]. Based on a Random Forest ensemble, Marin et al. propose a method to combine multiple local experts in order to accurately detect pedestrians [29]. Takarli et al. proposed detecting pedestrians using a combination of global and local features [30].

However, the traditional detection methods based on artificially-designed features to train the classifier do not work well on vehicles in a variety of complex real-world conditions, such as low light, rainy days, motion blur, different positions of the vehicle in the frame, and variations of the environment. For these reasons, they are not suitable for applications in automated vehicles or advanced driving assistance system.

3) DEEP NEURAL NETWORK BASED DETECTION METHODS

There are three main categories for object detection neural networks, namely: one-stage, two-stage and transformers. One-stage networks such as YOLO and SSD have predefined overlapping regions of the frame to detect and classify objects inside each region. A filtering process is performed to remove regions that are overlapping on one single object [31], [32]. On the other hand, two-stage networks, such as RCNN and Fast-RCNN, contain a pipeline to perform both the region proposal and classification of the regions [33], [34]. Comparatively, One- stage networks are generally faster, but will have lower accuracy compared to two-stage networks. Finally, vision transformers such as BERT or DETR, divide the frames into patches and then search for relationship between pixels, however they require a considerable amount of training data [35], [36].

In order to ensure accurate detection, many automotive algorithms have been based on two-stage detection model, with Fast R-CNN and Faster R-CNN being the most used. Nguyen has proposed an improved Faster-RCNN vehicle detection algorithm to address the problems of large-scale variation and mutual occlusion in vehicle detection, with a 4% performance improvement compared to Faster-RCNN [37]. Rui et al. have developed the Feature Pyramid Network (PRN), based on the Faster R-CNN, for pedestrian detection [38]. Zhang et al. have employed Faster R-CNN to implement pedestrian detection based on infrared images [39].

However, on-board object detection requires pressing real-time performance, and consequently, more and more one-stage models have been investigated for automotive applications. Since the first version of YOLO proposed by Redmon et al., there have been many refinements and improvements made, spanning several versions [31]. YOLOv3 is the most recent variant proposed by the original author [40]. YOLOv4 continues from the base framework from YOLOv3, incorporating optimisation and improvement methods such as mosaic data augmentation, mish activation function and dropblock [41]. YOLOv4 has shown to have an improvement in performance of 10% in average precision and 12% in speed (frames per second) compared to YOLOv3 in MS COCO dataset (test-dev 2017) [41]. Jamiya and Rani have addressed the difficulty of balancing the speed and accuracy of current vehicle detection algorithms by enhancing YOLOv3 and incorporating the concept of Spatial Pyramid Pooling [42]. The proposed YOLO-SPP detection algorithm has shown good real-time performance, allowing timely responses in vehicle's warning systems. Moreover, Chao et al. have achieved an enhancement of detection of overlapped targets using the SSD algorithm and adding a rejection term to the DNN loss function [43]. There are further variants of YOLO from various groups, building on the spirit of the YOLO network, increasing performance in speed and accuracy by tweaking the architectures and incorporating new features [44], [45].

C. CAMERA COLOUR PIPELINE AND DNNs

Some researchers have also started to investigate the possible impact of the quality/type of input data on the DNN performance.

Liu et al. examined the effect of camera parameters on the neural network and experimentally demonstrated that there was little difference between the detection of vehicles with monochrome and RGB frames when using Mask R-CNN as the detection algorithm [20]. Some groups have investigated the effects of ISP-processed frames on DNN detection, but the core of the work is not focused on automotive specific datasets and tasks. Hansen et al. have 'inverted' an ISP pipeline, considering a few building blocks of a generic ISP, as shown in Fig. 2b [46]. However, the Authors acknowledge that ISP is not invertible and therefore their inversion might add or modify the information contained in the original frames in unexpected ways. The ISP-processed frames perform better than the 'inverted' raw frames, according to the DNN used, however it is not clear how the 3 channel input network is adapted to use the 'raw' data. Hansen et al. also present an ablation study based on the ISP blocks considered in their ISP model, stating that each block contributes to a performance enhancement for the DNN, except for the denoising. The Tone Mapping is reported as the block most beneficial to the accuracy of detection. The Authors also re-converted the 'inverted' raw into 'simulated' RGB frames, and in this case DNN performance are still lower than on the original dataset [46]. This result highlights the need of investigating more the 'inversion' process before considering the achieved results reliable and generalisable. Lubana et al. propose a simplified version of ISP by selecting some arbitrary blocks in the colour pipeline and evaluating the detection of processed frames on a trained deep neural network [47]. The results of their proposed algorithm show that frame detection after their proposed algorithm is better than on raw frames. However, the raw dataset used is very small (*i.e.* 225 frames) and the DNN input architecture and training is not fully described. The full architecture is not described either in the recent paper by Cahill et al., but interestingly they use one-stage and two-stage object detector to evaluate different types of Bayer data, focusing on gamma-correction [48]. However, the work only compares results of the DNN trained and tested on the same type of data, and the choice of gamma-correction is not fully convincing as a key step in the ISP pipeline [46]. Finally, recent work has investigated also end-to-end DNN methods to substitute the ISP pipeline, traditionally optimised for human vision, showing that for example these methods can outperform traditional ISP pipeline in the case of low light conditions and for machine learning applications [49], [50]. These works demonstrate once more that image processing has been traditionally optimised for human vision, and better looking frames do not necessarily produce the best results in the case of machine learning application.

TABLE 1. Conversion processes for creating the different versions of the dataset, where subscripts indicate pixel position in an original 2 × 2 block, bold lines are used to contour each pixel (with one or three intensity values), dotted line split colour channels of one pixel. CFA stands for Colour Filter Array, G_{av} represents the average of the $G_{1,2}$ and $G_{2,1}$ green pixels.

Format Number	Format	Colour filter array	Comments
1	Original RGB	$R_{1,1}$ $G_{1,1}$ $B_{1,1}$ $R_{1,2}$ $G_{1,2}$ $B_{1,2}$ $R_{2,1}$ $G_{2,1}$ $B_{2,1}$ $R_{2,2}$ $G_{2,2}$ $B_{2,2}$	Original non modified KITTI frames, with **three colour channels** (3 colour values per each pixel)
2	Grayscale	$Gray_{1,1}$ $Gray_{1,2}$ $Gray_{2,1}$ $Gray_{2,2}$	This format is derived from the original dataset by applying a grayscale algorithm, resulting in a **single colour channel**
3	Gray Bayer	$R_{1,1}$ $G_{1,2}$ $G_{2,1}$ $B_{2,2}$	This format is composed by selecting only one colour channel (*i.e.* only one intensity) per pixel from 1) assuming an RGGB CFA, resulting in a **single colour channel**
4	Bayer 0-filled	$R_{1,1}$ 0 0 0 $G_{1,2}$ 0 0 $G_{2,1}$ 0 0 0 $B_{2,2}$	This format is composed by keeping an intensity value in a pixel for each channel only if it corresponds to the correct colour and position based on a RGGB CFA. Other pixels values are set to 0, resulting in **three colour channels**
4b	Bayer 0-filled (GRBG)	0 $G_{1,1}$ 0 $R_{1,2}$ 0 0 0 0 $B_{2,1}$ 0 $G_{2,2}$ 0	This format is a variant of 4, assuming GRBG CFA and resulting in **three colour channels**
4c	Bayer 0-filled (GBRG)	0 $G_{1,1}$ 0 0 0 $B_{1,2}$ $R_{2,1}$ 0 0 0 $G_{2,2}$ 0	This format is a minor variant of 4, with a GBRG CFA and resulting in **three colour channels**
4d	Bayer 0-filled (BGGR)	0 0 $B_{1,1}$ 0 $G_{1,2}$ 0 0 $G_{2,1}$ 0 $R_{2,2}$ 0 0	This format is a minor variant of 4, with a BGGR CFA and resulting in **three colour channels**
5	Bayer colour-filled	$R_{1,1}$ G_{av} $B_{2,2}$ $R_{1,1}$ $G_{1,2}$ $B_{2,2}$ $R_{1,1}$ $G_{2,1}$ $B_{2,2}$ $R_{1,1}$ G_{av} $B_{2,2}$	This format is composed by keeping an intensity value in a pixel for each channel only if it corresponds to the right colour and right position based on a RGGB CFA. For each 2x2 block, in the red and blue channels, the selected value is replicated in the other pixels. For the green channel, an average of the two green values is used to fill the 2 empty green values. **Three colour channels**

Furthermore, there are several recent studies optimising ISP selection and parameters for machine learning based perception [51], [52], [53]. However, they do not provide the benefit of reduction in processing and data rate compared to directly using raw data.

III. METHODOLOGY

As previously stated, the processing introduced in cameras to convert a single value channel 'frame' into three colour channels has been created for human vision. This work investigates if *Bayer* information can be used for object detection by machine vision without degrading the detection performance with respect to traditional RGB frames. The following subsections explain the steps of our methodology, Fig. 3, and particularly our methods to convert an existing automotive dataset into *Bayer* frames, allowing the ground truth bounding box information to be retained.

A. DATASET

Most commonly used automotive datasets generally provide three colour channel frames captured from automotive cameras. A recently released dataset by Oxford University, the

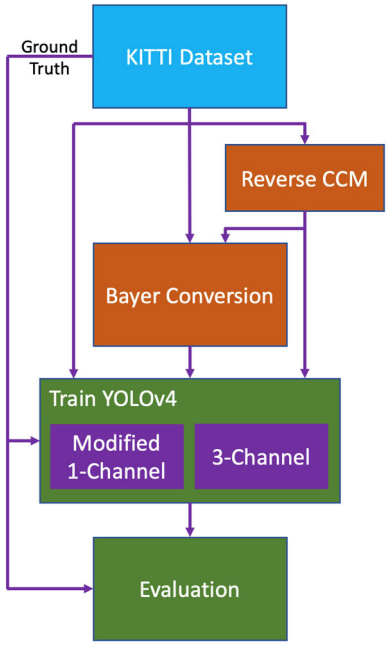


FIGURE 3. Flow diagram showing the methodology followed for the presented experiments. 11 different variants of the dataset have been created and then used as training or testing data using YOLOv4 DNN.



FIGURE 4. Original detail from a KITTI frame top left, Bayer 0-Filled frame top right, Bayer Colour-Filled bottom left and single channel gray Bayer bottom right.

RobotCar dataset, contains unrectified, 8-bit single colour channel Bayer frames, top frame in Fig. 1 [17]. However, this dataset does not provide labelled data. To the aim of this work, an automotive benchmarking dataset, the KITTI dataset, was chosen for the experiments and converted into Bayer frames [8]. The different methods to convert the dataset into Bayer are explained in Sec. III-B. In total, 8 three-channel datasets were generated (*i.e.* based on two different methods and four variants from colour correction matrix and colour filter array alignment), and 3 single-channel datasets.

B. CONVERSION OF DATASET

The selected KITTI dataset provides post-processed frames which have been through the used camera image signal processing (ISP) and frame rectification. ISP processing is not fully reversible and the specific ISP pipeline has not been released. In the work by Hansen et al., the Authors have



FIGURE 5. Detail from a frame from the KITTI dataset: left is the original frame, and right is the same frame with the CCM inverse applied to it.

tried to revert the ISP pipeline starting from RGB frames, and then apply again the ISP process to create 'simulated' RGB frames [46]. However these 'simulated' frames had different performance with respect to the original RGB. In the hereby presented work, to avoid further modifications to the pre-processed data, we have created our *Bayer* datasets using as much as possible the values stored in the frames of the original KITTI dataset (from now on named 'Original RGB' dataset). We have also investigated and validated this approach by investigating the placement of the colour filter array (the CFA configuration is not known *a priori*). The different formats of the generated *Bayer* datasets are listed in Table 1 and described below.

In Table 1, there are two *single channel formats* consisting of grayscale and Gray Bayer. The produced Gray Bayer dataset, format 3, uses the colour channel value based on the colour filter for each pixel, similar to how in cameras the CFA creates one intensity value in each pixel. Format 2 dataset, grayscale, was created using a grayscaling algorithm which interpolates using the RGB values for every pixel and is used to act as a comparison against the generated Gray Bayer. In the case of single channel inputs we needed to modify YOLOv4, as explained in Sec. III-D.

To allow a comparison of *Bayer* performance against RGB frames, three-colour channel *Bayer* frames were created to use the neural network without modification. These three-colour channel *Bayer* frame formats are designed to not modify the information content in the Bayer frames. Format 4, Bayer 0-Filled, contains the same values as format 3, Gray Bayer, but split into the correct colour channels with the remaining pixels being filled with zero value. Although the information has not changed, the introduction of the zeros might have an implication on the neural network detection (due to the zero patterns in the three colour channels). Hence, a second three channel *Bayer* frame format dataset, Bayer Colour-Filled (format 5), was also generated. In this format, the pixel channels without values are instead filled with the corresponding pixel value of that channel in the 2×2 matrix, except green where it is filled with an average of the two values in the 2×2 matrix. Fig. 4 shows a detail from a frame of the KITTI dataset to visually compare a frame generated with formats 3 to 5 and used in the presented

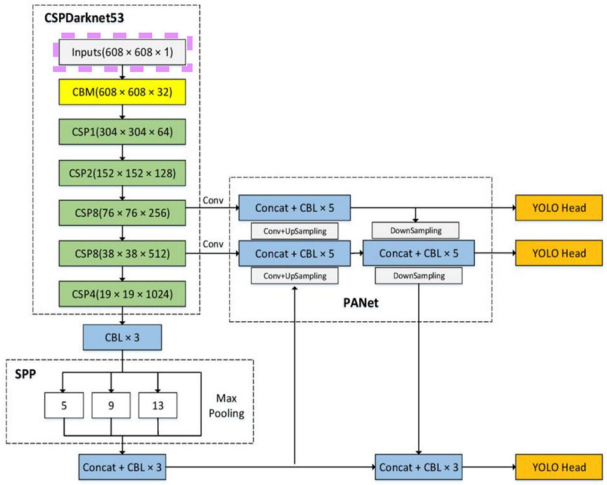


FIGURE 6. YOLOv4 architecture with the input layer (highlighted with dotted purple box) modified to accept one channel input.

experiments. As human consumers, the original frame (top left) is the most pleasant frame, without 'abrupt' changes and with 'clear' details.

Moreover, the CFA placement is considered in this work. The alignment of the CFA for the KITTI dataset is not known. This work tests the different possible alignments of the CFA (*i.e.* RGGB, GRBG, GBRG, BGGR) to understand if it will have an effect on the evaluation of *Bayer* frame with the selected neural networks. Three additional variants of the Bayer 0-Filled format were created based on the other alignment of the CFA, see formats 4b to 4d in Table 1.

C. COLOUR CORRECTION MATRIC (CCM)

One critical part of the ISP is to perform colour correction through a colour correction matrix (CCM), generating a more natural frame to the human visual system [46]. To investigate

$$CCM = \begin{bmatrix} 1.6605 & -0.5876 & -0.0728 \\ -0.1246 & 1.1329 & -0.0083 \\ -0.0182 & -0.1006 & 1.1187 \end{bmatrix} \quad (1)$$

the effect of the CCM, a generic CCM, eq. 1, was inversed and applied to the dataset based on [54], see also Fig. 5. This process was performed to the original RGB dataset, Bayer 0-Filled and Gray Bayer which are the two formats most representative of a *Bayer* frame generated. The inverse CCM was applied on the original frame before the Bayer conversion was performed.

D. YOLOv4

A one-stage network, namely a YOLOv4, is chosen for this work due to strict real-time requirements of the functions deployed for assisted and automated driving, see. Sec II-B. Several recent works have demonstrated that in the case of KITTI and automotive datasets, trends observed in one- and two-stage detectors are similar [48], [55]. As a consequence, to reduce variability, our work focuses on one architecture, but creating 11 trained DNN versions (see below) and cross-evaluate several datasets (*i.e.* original and different versions of one or three colour channel *Bayer*) per trained network. Our methodology can be followed and applied to any DNN architecture. To the best of our knowledge, this is the first study of this type. The input of the selected network requires RGB (3 channels) frames, so as a part of this work a one channel input version of the YOLOv4 was created, but only the input layer of the network was modified, see Fig. 6. Part of the experiments are carried out with the DNN version with one channel input and part with three channels input, the generation of input data from the original RGB data is described in Table 1. On the contrary, previous studies feeding 'Bayer' data to DNN do not specify how the 'Bayer' data nor the DNNs were modified, so they are not fully reproducible.

As a part of this work, the selected DNN was re-trained with different datasets, as described in Sec III. A-C, generating 11 versions of the re-trained network, of which 8 three channels input (*i.e.* original RGB, 0-filled Bayer, Colour-filled Bayer, original RGB no CCM, 0-filled Bayer no CCM, and 0-filled Bayer with 3 more variations of the CFA) and 3 one channel input (*i.e.* grayscale, gray Bayer, gray Bayer no CCM). These re-trained networks were then used to evaluate different three channel or one channel datasets, depending on the specific experiment. All combinations between training and testing are reported in Tables 2-4. The initial YOLO model parameters were initialised with small random numbers, close to 0. The model parameters were updated based on category loss, bounding box regression loss, and target confidence loss through 40 training epochs for every trained network.

E. EVALUATION

The evaluation metrics selected in this paper are based on mean average precision, *i.e.* $mAP_{0.5}$ and $mAP_{[0.5:0.95]}$. For $mAP_{0.5}$, the mean average precision across the classes is calculated, given an intersection over unit (IoU), between the predicated bounding box and ground truth of 0.5. In $mAP_{[0.5:0.95]}$, the mAP is computed for each step of IoU between 0.5 and 0.95, with a step size of 0.05, and is then averaged. In the automotive field, $mAP_{0.5}$ show that objects are identified correctly, but can have a higher degree of uncertainty in the location of the object. On the other hand, $mAP_{[0.5:0.95]}$ evaluates with multiple steps of increasing IoUs, thus the score also considers how accurate is the location of the identified object in the frame, which can be key for the safety of assisted and automated driving functions.

IV. RESULTS

A total of 41 different pairings of the 11 re-trained DNNs and 11 generated datasets have been evaluated, of which 33 pairs related to the three channel input DNN and 8 pairs to the one channel input DNN version. Of the three single channel re-trained networks, the gray Bayer was cross-evaluated with respect to the grayscale (generating 4 sets of results) and with respect to the gray Bayer with no CCM (further 4 sets of results). For the 3 channel inputs, the Original, Bayer 0-filled

TABLE 2. Table of results for the differently trained networks evaluated with the different datasets. Data formats are explained in Table 1.

Network Input Channels	Training set type	Evaluation set type	$mAP_{0.5}$	$mAP_{[0.5:0.95]}$
YOLOv4 with three channel input	Original RGB	Original RGB	**0.915**	**0.558**
		Bayer 0-Filled	0.828	0.507
		Bayer Colour-Filled	0.895	0.534
	Bayer 0-Filled	Original RGB	0.876	0.508
		Bayer 0-Filled	**0.897**	**0.522**
		Bayer Colour-Filled	0.876	0.506
	Bayer Colour-Filled	Original RGB	**0.912**	**0.552**
		Bayer 0-Filled	0.657	0.379
		Bayer Colour-Filled	**0.912**	0.549
YOLOv4 with one channel input	Grayscale	Grayscale	**0.907**	0.541
		Gray Bayer	0.903	**0.542**
	Gray Bayer	Grayscale	0.884	0.524
		Gray Bayer	**0.909**	**0.542**

and colour-filled were cross-evaluated (generating 9 sets of results); the 4 different configurations of CFA were cross-evaluated (generating 16 sets of results); and also Original and 0-filled versions were cross-evaluated with respect their inversion using the CCM (generating 8 more sets of results). All the results are reported in Tables 2-4.

A. QUALITATIVE EVALUATION

Two adequately different frames were selected from the dataset to show the detection results, *i.e.* 000232.png and 000400.png, they are shown side by side in Fig. 7-8, where the detections when the DNN are trained and tested with the same variant of the dataset. The selected frames are different in terms of visual content: Fig. 7. is not crowded but has 3 different types of road stakeholders, including one vulnerable stakeholder, *i.e.* the bike. Fig. 8. has several vehicles of different sizes and with different levels of occlusion. The detections and classifications with confidence scores for different objects are overlayed on the frames. Overall the detections look very similar in all the selected cases.

TABLE 3. Evaluation results comparing different Colour Filter Array Placement (Table 1) when generating the Bayer 0-Filled.

Training Set	Evaluation Set Bayer 0-filled	$mAP_{[0.5]}$	$mAP_{[0.5:0.95]}$
Bayer 0-filled (4)	RGGB (4)	**0.897**	**0.522**
	GRBG (4b)	0.887	0.510
	GBRG (4c)	0.891	0.517
	BGGR (4d)	**0.897**	0.513
Bayer 0-filled GRBG (4b)	RGGB (4)	0.894	0.521
	GRBG (4b)	0.891	0.520
	GBRG (4c)	0.889	**0.529**
	BGGR (4d)	**0.897**	0.524
Bayer 0-filled GBRG (4c)	RGGB (4)	0.895	0.535
	GRBG (4b)	0.895	0.528
	GBRG (4c)	0.890	0.517
	BGGR (4d)	**0.899**	**0.537**
Bayer 0-filled BGGR (4d)	RGGB (4)	0.887	0.527
	GRBG (4b)	0.886	**0.531**
	GBRG (4c)	0.884	0.518
	BGGR (4d)	**0.893**	0.524

B. QUANTITATIVE EVALUATION

The main results have been split into three tables (Tables 2-4) to identify three main aspects: comparing the detection performance of the DNNs trained with different types of data when evaluating RGB versus *Bayer* data; understanding the role of the position of the colour filter array on the results; analysing the role of the colour correction matrix. The top half of Table 2 shows the results in terms of the selected metrics, *i.e.* $mAP_{0.5}$ and $mAP_{[0.5:0.95]}$, when the YOLOv4 network accepts a three colour channel input, and the bottom half presents the results of the modified YOLOv4 accepting only a single colour channel input. For three channel input version, YOLO has been re-trained with original RGB data, Bayer 0-filled, and Bayer colour-filled and cross-evaluated across these formats, for the one input the network has been re-trained with Grayscale and Gray Bayer and cross-evaluated. The highest metrics values for each trained network have been highlighted in bold, and the best metrics across the different combinations show comparable performance within 5%.

Table 3 presents the results of the DNN re-trained with the Bayer 0-filled considering 4 different RGB CFA placement, as shown in Table 1 (*i.e.*4, 4b, 4c, 4d). The 4 generated three channel networks have been cross-evaluated with testing data generated with the 4 CFA placements too. The highest metrics values for each trained network have been highlighted in bold, and the best metrics across the different combinations show comparable performance within 2%, with $mAP_{0.5}$ varying of less than 1%.

TABLE 4. Evaluation results comparing the removal of a pseudo CCM.

Training Set Type	Evaluation Set Type	$mAP_{[0.5]}$	$mAP_{[0.5:0.95]}$
Original RGB	Original RGB	**0.915**	**0.558**
	Original RGB No CCM	0.912	0.555
Original RGB No CCM	Original RGB	0.913	0.554
	Original RGB No CCM	**0.916**	**0.562**
Bayer 0-Filled	Bayer 0-Filled	**0.897**	**0.522**
	Bayer 0-Filled No CCM	0.877	0.511
Bayer 0-Filled No CCM	Bayer 0-Filled	**0.902**	**0.540**
	Bayer 0-Filled No CCM	0.894	0.533
Gray Bayer	Gray Bayer	**0.909**	**0.542**
	Gray Bayer No CCM	0.874	0.512
Gray Bayer No CCM	Gray Bayer	0.893	**0.533**
	Gray Bayer No CCM	**0.897**	0.528

Finally, Table 4 compares the results when trying to remove the effect of the CCM processing from the original RGB data. In this case 4 different three input channel networks have been re-trained with the original RGB data, original RGB before CCM (cross-evaluating these two formats), Bayer 0-filled and Bayer 0-filled without CCM (again cross-evaluating these two formats), and then 2 different one input channel networks have been trained with Gray Bayer and Gray Bayer before CCM (cross-evaluating too). The highest metrics values for each trained network have been highlighted in bold, and the best metrics across the different combinations show comparable performance within 2%, with $mAP_{0.5}$ varying of less than 1%.

V. DISCUSSION

In terms of the qualitative results (Figs. 7-8), the detections for the selected pairs training-testing are extremely similar in the two frames, with small variations of the confidence scores. An interesting aspect is that even in the case of occluded vehicles and vulnerable road users (*e.g.* the cyclist), the DNN is able to classify them correctly for all the data formats used. Overall the detection of vulnerable road users

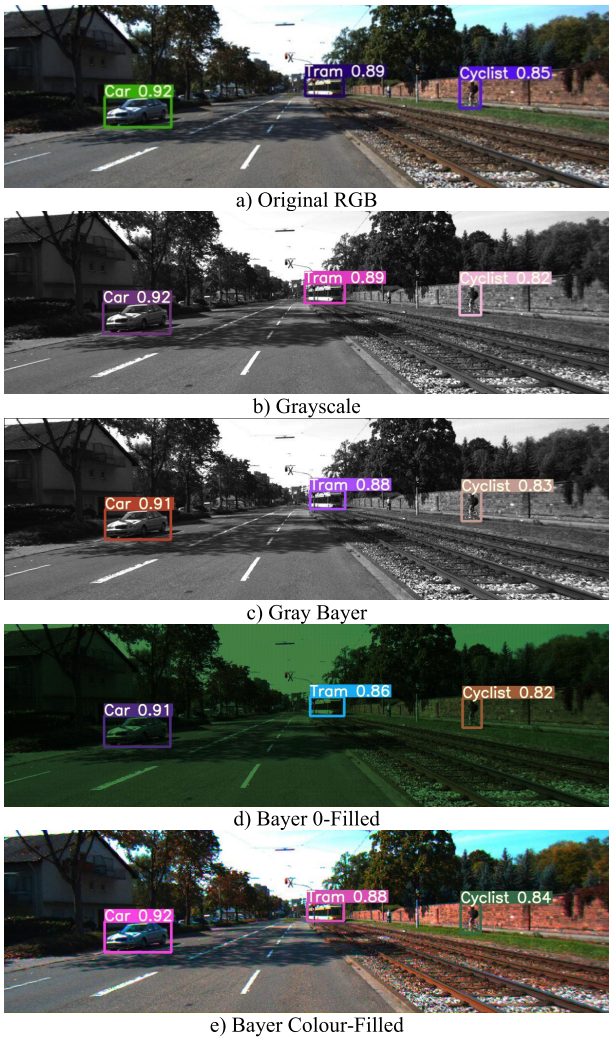


FIGURE 7. Objects detected by the trained network overlayed onto the frame 000232.png. The training and evaluating dataset uses the same frame formats.

and small objects in the frame does not seem detrimentally affected when using the different formats of Bayer data, and performance are very close to the performance of the original RGB data.

These results are further confirmed by the values reported in the Tables. 2-4. In all the training-testing combinations, the best performance with each network trained with a different variant of the dataset ranged between 0.893 to 0.916 for $mAP_{0.5}$, and 0.522 to 0.568 for $mAP_{[0.5:0.95]}$. These results suggest that when using DNN based object detectors, there are minor performance variations in the 'detection' using different representation of the data. However, the accuracy of the bounding boxes (*i.e.* position and size) may suffer slightly more. On a high level, the different ways of representing Bayer information in Table 2 contained the same information, derived from the original dataset, but are arranged differently. Hence, the achieved values demonstrate that the DNN can cope with small changes in how the information is fed to the network. However, it seems that the Bayer 0-filled version yields to the worst performance (*i.e.* 2% lower than



FIGURE 8. Objects detected by the trained network overlayed onto the frame 000400.png. The training and evaluating dataset uses the same frame formats.

the RGB-RGB training/evaluation version), this performance decrease might be due to zeros pattern in the input data hindering the network convolution and feature extraction. The Bayer colour-filled trained network has a very interesting property: the detection performance in terms of $mAP_{0.5}$ is the same when evaluating the Bayer colour-filled and the original RGB data and only 0.3% different from the RGB-RGB DNN performance. It means that actually the performance when using this format of data are indistinguishable from traditional RGB based DNNs, but also that hyperparameter tuning can be implemented for the colour-filled Bayer re-trained YOLO, yielding to even higher performance, further enhancing $mAP_{[0.5:0.95]}$. These results imply that state-of-the-art networks can be re-used and be further optimised for consumption of Bayer data. This aspect will enable an immediate reduction of the bandwidth required for camera data transmission on traditional vehicle communication networks, in the sense that camera data can be transmitted as non-processed single channel Bayer, and then 'colour-filled' to three channels in the DNN input stage. Moreover, recent work has mentioned that the use of raw frames can reduce overall sensor power consumption (up to 35%) and the processing time of one sixth of the framerate, so the use of Bayer frames in automotive can bring a significant optimisation when using cameras for assisted and automated driving functions [48].

For the network adapted for single channel input, using greyscale frame and gray Bayer frames performed extremely similarly. However, the grey Bayer (training and evaluation) DNN performed the best for both $mAP_{[0.5]}$ and $mAP_{[0.5:0.95]}$. Additionally, YOLO trained and evaluated with grey Bayer performed very similarly to the network trained and evaluated with the original RGB, with a negligible decrease of 0.6% in $mAP_{0.5}$ and of 1.6% in $mAP_{[0.5:0.95]}$. These results show that single channel Bayer frames can be a promising direction for research, with possible performance gain through slimmer networks (smaller inputs), optimised network architectures, and hyperparameter tuning.

In addition, as mentioned earlier, the exact colour filter array placement in the frame is not known. This issue could affect the conversion of the dataset, see 4 to 4d formats in Table 1. To consider how the exact CFA placement affect the DNN output, we have trained and cross-evaluated YOLO with the four CFA datasets. The results are recorded in Table 3. Due to the demosaicing process used to produce the RGB frame, every pixel channel has a dependence and relation to neighbouring pixels. Hence, although a different CFA pattern is applied, there are no major differences in pixel value patterns and frame features should still be recognisable. Therefore, the colour filter array orientation does not play a large role in this work.

Finally, Table 4 compares some of the best performing data formats with their version pre-CCM. For the original RGB data, CCM do not seem to play a significant role in the performance. This is similar for the Bayer 0-filled and the Gray Bayer. This outcome shows once more that the processing in the ISP is not really optimised for DNN perception, and therefore it can be removed whereas more effort is allocated into converting existing DNNs to use Bayer data and maximising their performance.

VI. CONCLUSION

Building on the research and automotive trend of linking raw data with perception tasks, this paper presents a study on the use and re-training of DNN based object detectors to consume Bayer data instead of traditional RGB data, without any modifications to the neural network architecture. Moreover, with minimal adjustments, a DNN has been also converted to use single channel frames, and when using the Gray Bayer dataset, the DNN performance have been almost identical (with a variation of 0.6%) to the traditional version of the Neural Network. The placement of CFA on sensors and the role of CCM have been analysed and discussed, and overall their effect seems marginal for the network performance. In alignment with recent work, the achieved results show that whilst the internal processing on camera sensors has been optimised for human vision, most of the implemented

processing is not really needed for DNN based perception, and specifically for object detection. These findings not only pave the way for the re-use of current DNN in order to consume Bayer data, but also open the possibility to develop optimised architectures to use Bayer. In turns these achievements can improve the safety of future assisted and automated driving functions and also their efficiency, in terms of less sensor data to be transmitted to the vehicle processing unit(s), less sensor power consumption, and reduction of latency due to ISP. Furthermore, for real-world deployments including infrastructure sensing or off-board processing (*i.e.* edge processing), the use of Bayer data can reduce the bandwidth pressure that is faced due to communication channel limitations. Future work will explore other downstream perception tasks and datasets. Additionally, how different DNN architectures for Bayer data can be explored and developed to best optimise the performance of the detection consuming this form of input data.

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REFERENCES

[1] M. Cococcioni, F. Rossi, E. Ruffaldi, S. Saponara, and B. D. de Dinechin, "Novel arithmetics in deep neural networks signal processing for autonomous driving: Challenges and opportunities," *IEEE Signal Process. Mag.*, vol. 38, no. 1, pp. 97–110, Jan. 2021.

[2] *Taxonomy and Definitions for Terms Related to Driving Automation Systems for On-Road Motor Vehicles*, Standard J3016_202104, SAE, 2021.

[3] H. Zhu, K.-V. Yuen, L. Mihaylova, and H. Leung, "Overview of environment perception for intelligent vehicles," *IEEE Trans. Intell. Transp. Syst.*, vol. 18, no. 10, pp. 2584–2601, Oct. 2017, doi: 10.1109/TITS.2017.2658662.

[4] H. Guo, Y. Zhang, S. Cai, and X. Chen, "Effects of level 3 automated vehicle drivers' fatigue on their take-over behaviour: A literature review," *J. Adv. Transp.*, vol. 2021, pp. 1–12, Oct. 2021.

[5] T. Cohen and C. Cavoli, "Automated vehicles: Exploring possible consequences of government (non)intervention for congestion and accessibility," *Transp. Rev.*, vol. 39, no. 1, pp. 129–151, Jan. 2019.

[6] M. Abbasi, A. Shahraki, and A. Taherkordi, "Deep learning for network traffic monitoring and analysis (NTMA): A survey," *Comput. Commun.*, vol. 170, pp. 19–41, Mar. 2021.

[7] Z. Zou, K. Chen, Z. Shi, Y. Guo, and J. Ye, "Object detection in 20 years: A survey," *Proc. IEEE*, vol. 111, no. 3, pp. 257–276, Mar. 2023, doi: 10.1109/JPROC.2023.3238524.

[8] A. Geiger, P. Lenz, C. Stiller, and R. Urtasun, "Vision meets robotics: The KITTI dataset," *Int. J. Robot. Res.*, vol. 32, no. 11, pp. 1231–1237, Sep. 2013.

[9] Z. Zheng, L. Yang, Y. Wang, M. Zhang, L. He, G. Huang, and F. Li, "Dynamic spatial focus for efficient compressed video action recognition," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 2, pp. 695–708, Feb. 2024.

[10] C.-P. Hsu, B. Li, B. Solano-Rivas, A. R. Gohil, P. H. Chan, A. D. Moore, and V. Donzella, "A review and perspective on optical phased array for automotive LiDAR," *IEEE J. Sel. Topics Quantum Electron.*, vol. 27, no. 1, pp. 1–16, Jan. 2021, doi: 10.1109/JSTQE.2020.3022948.

[11] P. H. Chan, A. Huggett, G. Souvalioti, P. Jennings, and V. Donzella, "Influence of AVC and HEVC compression on detection of vehicles through faster R-CNN," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 1, pp. 203–213, Jan. 2024, doi: 10.1109/TITS.2023.3308344.

[12] X. Dong, Q. Guo, T. Gan, Q. Wang, J. Wu, X. Ren, Y. Cheng, and W. Chu, "SNP-S^3: Shared network pre-training and significant semantic strengthening for various video-text tasks," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 4, pp. 2525–2535, Apr. 2024, doi: 10.1109/TCSVT.2023.3303945.

[13] Y. Pan, C. Ren, X. Wu, J. Huang, and X. He, "Real image denoising via guided residual estimation and noise correction," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 4, pp. 1994–2000, Apr. 2023, doi: 10.1109/TCSVT.2022.3216681.

[14] B. Song, J. Zhou, X. Chen, and S. Zhang, "Real-scene reflection removal with RAW-RGB image pairs," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 8, pp. 3759–3773, Aug. 2023, doi: 10.1109/TCSVT.2023.3241319.

[15] Y. Fu, Y. Hong, Y. Zou, Q. Liu, Y. Zhang, N. Liu, and C. Yan, "Raw image based over-exposure correction using channel-guidance strategy," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 4, pp. 2749–2762, Apr. 2024.

[16] R. Xu, C. Chen, J. Peng, C. Li, Y. Huang, F. Song, Y. Yan, and Z. Xiong, "Toward RAW object detection: A new benchmark and a new model," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Vancouver, BC, Canada, Jun. 2023, pp. 13384–13393, doi: 10.1109/cvpr52729.2023.01286.

[17] W. Maddern, G. Pascoe, C. Linegar, and P. Newman, "1 year, 1000 km: The Oxford RobotCar dataset," *Int. J. Robot. Res.*, vol. 36, no. 1, pp. 3–15, Jan. 2017.

[18] K. Chang, H. Li, Y. Tan, P. L. K. Ding, and B. Li, "A two-stage convolutional neural network for joint demosaicking and super-resolution," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 7, pp. 4238–4254, Jul. 2022, doi: 10.1109/TCSVT.2021.3129201.

[19] P. H. Chan, G. Souvalioti, A. Huggett, G. Kirsch, and V. Donzella, "The data conundrum: Compression of automotive imaging data and deep neural network based perception," in *Proc. London Imag. Meeting*, vol. 2, Sep. 2021, pp. 78–82.

[20] Z. Liu, T. Lian, J. Farrell, and B. A. Wandell, "Neural network generalization: The impact of camera parameters," *IEEE Access*, vol. 8, pp. 10443–10454, 2020.

[21] A. Bensrhair, M. Bertozzi, A. Broggi, P. Miche, S. Mousset, and G. Toulminet, "A cooperative approach to vision-based vehicle detection," in *Proc. IEEE Intell. Transp. Syst.*, May 2001, pp. 207–212.

[22] J. M. Collado, C. Hilario, A. de la Escalera, and J. M. Armingol, "Model based vehicle detection for intelligent vehicles," in *Proc. IEEE Intell. Vehicles Symp.*, Aug. 2004, pp. 572–577.

[23] Y. Chong, W. Chen, Z. Li, W. H. K. Lam, C. Zheng, and Q. Li, "Integrated real-time vision-based preceding vehicle detection in urban roads," *Neurocomputing*, vol. 116, pp. 144–149, Sep. 2013.

[24] R. A. Hadi, G. Sulong, and L. E. George, "Vehicle detection and tracking techniques: A concise review," *Signal Image Process., Int. J.*, vol. 5, no. 1, pp. 1–12, Feb. 2014.

[25] A. Haselhoff and A. Kummert, "A vehicle detection system based on Haar and triangle features," in *Proc. IEEE Intell. Vehicles Symp.*, Jun. 2009, pp. 261–266.

[26] N. Laopracha, T. Thongkrau, K. Sunat, P. Songrum, and R. Chamchong, "Improving vehicle detection by adapting parameters of HOG and kernel functions of SVM," in *Proc. Int. Comput. Sci. Eng. Conf. (ICSEC)*, Jul. 2014, pp. 372–377.

[27] X. Cao, C. Wu, P. Yan, and X. Li, "Linear SVM classification using boosting HOG features for vehicle detection in low-altitude airborne videos," in *Proc. 18th IEEE Int. Conf. Image Process.*, Sep. 2011, pp. 2421–2424, doi: 10.1109/ICIP.2011.6116132.

[28] B. T. Bastian, "Pedestrian detection using first- and second-order aggregate channel features," *Int. J. Multimedia Inf. Retr.*, vol. 8, no. 2, pp. 127–133, Jun. 2019.

[29] J. Marín, D. Vázquez, A. M. López, J. Amores, and B. Leibe, "Random forests of local experts for pedestrian detection," in *Proc. IEEE Int. Conf. Comput. Vis.*, Dec. 2013, pp. 2592–2599.

[30] F. Takarli, A. Aghagolzadeh, and H. Seyedarabi, "Combination of high-level features with low-level features for detection of pedestrian," *Signal, Image Video Process.*, vol. 10, no. 1, pp. 93–101, Jan. 2016.

[31] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Las Vegas, NV, USA, Jun. 2016, pp. 779–788, doi: 10.1109/CVPR.2016.91.

[32] W. Liu, D. Anguelov, D. Erhan, C. Szegedy, S. Reed, C.-Y. Fu, and A. C. Berg, "SSD: Single short MultiBox detector," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, vol. 9905, 2016, pp. 21–37.

[33] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich feature hierarchies for accurate object detection and semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Columbus, OH, USA, Jun. 2014, pp. 580–587, doi: 10.1109/CVPR.2014.81.
[34] R. Girshick, "Fast R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Santiago, Chile, Dec. 2015, pp. 1440–1448, doi: 10.1109/ICCV.2015.169.
[35] J. Devlin, M. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. Conf. North Amer. Chapter Assoc. Comptutational Linguistics, Hum. Lang. Technol.*, Jan. 2019, pp. 4171–4186.
[36] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Jan. 2020, pp. 213–229.
[37] H. Nguyen, "Improving faster R-CNN framework for fast vehicle detection," *Math. Problems Eng.*, vol. 2019, no. 1, Jan. 2019, Art. no. 3808064, doi: 10.1155/2019/3808064.
[38] T. Rui, J. Fei, Y. Zhou, H. Fang, and J. Zhu, "Pedestrian detection based on deep convolutional neural network," *Comput. Eng. Appl.*, vol. 52, no. 13, pp. 162–166, 2016.
[39] Z. Li-liang, L. Lin, X. Liang, and K. He, "Is faster R-CNN doing well for pedestrian detection?" in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Jan. 2016, pp. 443–457.
[40] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," 2018, *arXiv:1804.02767*.
[41] A. Bochkovskiy, C.-Y. Wang, and H.-Y. Mark Liao, "YOLOv4: Optimal speed and accuracy of object detection," 2020, *arXiv:2004.10934*.
[42] S. S. Jamiya and P. E. Rani, "LittleYOLO-SPP: A delicate real-time vehicle detection algorithm," *Optik*, vol. 225, Jan. 2021, Art. no. 165818, doi: 10.1016/j.ijleo.2020.165818.
[43] J. Cao, C. Song, S. Song, S. Peng, D. Wang, Y. Shao, and F. Xiao, "Front vehicle detection algorithm for smart car based on improved SSD model," *Sensors*, vol. 20, no. 16, p. 4646, Aug. 2020.
[44] C. Li, L. Li, H. Jiang, K. Weng, Y. Geng, L. Li, Z. Ke, Q. Li, M. Cheng, W. Nie, Y. Li, B. Zhang, Y. Liang, L. Zhou, X. Xu, X. Chu, X. Wei, and X. Wei, "YOLOv6: A single-stage object detection framework for industrial applications," 2022, *arXiv:2209.02976*.
[45] C.-Y. Wang, A. Bochkovskiy, and H.-Y. M. Liao, "YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors," 2022, *arXiv:2207.026696*.
[46] P. Hansen, A. Vilkin, Y. Krustalev, J. Imber, D. Talagala, D. Hanwell, M. Mattina, and P. N. Whatmough, "ISP4ML: The role of image signal processing in efficient deep learning vision systems," in *Proc. 25th Int. Conf. Pattern Recognit. (ICPR)*, Jan. 2021, pp. 2438–2445.
[47] E. S. Lubana, R. P. Dick, V. Aggarwal, and P. M. Pradhan, "Minimalistic image signal processing for deep learning applications," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2019, pp. 4165–4169.
[48] J. Cahill, A. Parsi, D. Mullins, J. Horgan, E. Ward, C. Eising, P. Denny, B. Deegan, M. Glavin, and E. Jones, "Exploring the viability of bypassing the image signal processor for CNN-based object detection in autonomous vehicles," *IEEE Access*, vol. 11, pp. 42302–42313, 2023, doi: 10.1109/ACCESS.2023.3270710.
[49] S. Diamond, V. Sitzmann, F. Julca-Aguilar, S. Boyd, G. Wetzstein, and F. Heide, "Dirty pixels: Towards end-to-end image processing and perception," *ACM Trans. Graph.*, vol. 40, no. 3, pp. 1–15, Jun. 2021.
[50] E. Tseng, A. Mosleh, F. Mannan, K. St-Arnaud, A. Sharma, Y. Peng, A. Braun, D. Nowrouzezahrai, J.-F. Lalonde, and F. Heide, "Differentiable compound optics and processing pipeline optimization for end-to-end camera design," *ACM Trans. Graph.*, vol. 40, no. 2, pp. 1–19, Apr. 2021.
[51] D. Molloy, B. Deegan, D. Mullins, E. Ward, J. Horgan, C. Eising, P. Denny, E. Jones, and M. Glavin, "Impact of ISP tuning on object detection," *J. Imag.*, vol. 9, no. 12, p. 260, Nov. 2023.
[52] Y. Guo, F. Luo, and X. We, "Learning degradation-independent representations for camera ISP pipelines," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 25774–25783.
[53] Z. Cui and T. Harada, "RAW-adapter: Adapting pre-trained visual model to camera RAW images," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, Aug. 2024, pp. 37–56.
[54] Z. Roman. *Color Correction With Matrix Transformation*. Accessed: Oct. 5, 2022. [Online]. Available: https://support.medialooks.com/hc/en-us/articles/360030737152-Color-correction-with-matrix-transformation
[55] B. Li, P. H. Chan, G. Baris, M. D. Higgins, and V. Donzella, "Analysis of automotive camera sensor noise factors and impact on object detection," *IEEE Sensors J.*, vol. 22, no. 22, pp. 22210–22219, Nov. 2022, doi: 10.1109/JSEN.2022.3211406.



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